Ben Sapirstein

Computational linguistics · typography · human-computer interaction

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I explore the meeting point of computer science, linguistics, and typography. I am interested in how the visual dimension of writing can shape understanding, and I develop approaches that treat the form of text itself as a source of information. In computational historical linguistics, I study how languages evolve and relate to one another, with a focus on the Semitic family.

EDUCATION

2022 – 2026 **M.Sc. Machine Learning & Data Science**, Reichman University, Herzliya
GPA 97/100. Thesis: SHAMS, an audio-grounded pronunciation benchmark for Levantine Arabic.

2017 – 2020 **B.Sc. Computer Science**, Hebrew University of Jerusalem
GPA 91/100. Algorithms, data science, deep learning, image processing, NLP, AI.

PUBLICATIONS

2026 **Sapirstein, B.**, Mattar, R., Mor-Lan, G., Mohamed, A., Cerqueglini, L., Alper, M. **SHAMS: An Audio-Grounded Pronunciation Benchmark for Levantine Arabic.** *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP @ EMNLP 2026)*.
arXiv:2610.01427 · shams-nlp.github.io

2025 **Sapirstein, B. Standardizing Phonetic Transcriptions for Kitchen et al.'s Comparative Wordlist on Semitic Languages with Language-Specific Orthography Profiles.** *Computer-Assisted Language Comparison in Practice*, 8(2), 95–105.
doi:10.15475/calcip.2025.2.5

2025 **Sapirstein, B. Pixels at BAREC Shared Task 2025: Visual Arabic Readability Assessment.** *Proceedings of the Third Arabic Natural Language Processing Conference: Shared Tasks*, 350–356.
ACL Anthology

RESEARCH

SHAMS. Levantine Arabic pronunciation benchmark: dataset, evaluation code, and project page.

TEACHING

Fall 2025 **Teaching Assistant, Computational Thinking and Programming in Python**, Reichman University
Weekly tutoring for undergraduates.

Fall 2025 **Course Assistant, Information Theory**, Reichman University

2020 – 2021 **Python Instructor**, Tech2Peace
Python tutor in a joint Israeli-Palestinian high-tech and peacebuilding seminar. Organized a hackathon with Madrasa in which teams built tools for teaching Palestinian dialectal Arabic to Hebrew speakers.

EXPERIENCE

2024 – 2026 **Data Scientist**, Microsoft, Herzliya
Fine-tuned large language models to autonomously generate security event detectors. Designed and implemented system workflows, analyzed sample data and reasoning outputs. Azure OpenAI.

2023 – 2024 **Algorithm Developer, Deep Learning**, Mobileye, Ramat Gan
Developed and trained real-time segmentation networks for autonomous vehicles, covering the pipeline from data engineering through model architecture to deployment. TensorFlow, Python, AWS, GCP.

SKILLS

Deep learning, computer vision, NLP · Python, C++, TensorFlow, AWS, GCP, Azure OpenAI
Linguistic data modeling, CLDF, comparative method, phonetic transcription

LANGUAGES

Hebrew (native) · English (fluent) · French (advanced) · Arabic (intermediate)